

Yash Saxena

✉ yash.saxena40@gmail.com ☎ (630) 219-9217 🔗 [linkedin.com/in/yashsaxena5](https://www.linkedin.com/in/yashsaxena5) 🐙 github.com/SpacelyProject

Technical Skills

Languages: Python, C/C++, TCL, SystemVerilog, Verilog, VHDL

ASIC Design & DV: RTL Architecture, UVM, Constrained-Random Verification, Assertions (SVA), STA/Timing

EDA Tools & Flow: Cadence Xcelium (xrun), Cocotb (Python), Vivado, ModelSim, MATLAB, Git

Lab Validation & Bring-Up: Cryogenic Testing (4K/77K), Spectrum/Phase Noise Analyzer, SMU, Oscilloscope, AWG, Power Optimization

Protocols & Interfaces: I2C, SPI, Wishbone, AXI4, SFP+ Optical Link

Professional Experience

ASIC Engineer

Batavia, IL

Fermilab

July 2023 – Present

- Designed digital readout RTL for the 32-channel SUNROCK cryogenic ASIC in SystemVerilog, implementing fine-delay TDCs with bubble-error filtering, multi-channel priority arbiters, and high-speed CML serializers.
- Implemented Triple Modular Redundancy (TMR) across on-chip configuration registers with real-time SEU counters for radiation tolerance; built dual slow-control interfaces using custom GPPI and an I2C-to-Wishbone bridge.
- Developed modular UVM testbenches for digital core blocks using Fermilab's *Songbird* framework, writing self-checking TLM scoreboards and constrained-random sequences to verify packet framing, FIFO backpressure, and CSR access.
- Developed AXI-based firmware on Zynq UltraScale+ (ZCU102) to capture serialized detector streams, directly correlating FPGA post-silicon bring-up with pre-silicon Cocotb and Xcelium simulations.
- Debugged 4K PLL frequency-lock failures on silicon by tuning VCO sub-bands and loop parameters, restoring stable lock with 1.875 ps RMS jitter; validated SPROCKET (SAR ADC for Skipper CCD) down to 77K.
- Built *Spacely*, an open-source Python framework (1000+ LOC) for automated SMU/AWG instrument control and wafer-level transistor characterization, reducing testing cycle time by ~80%.

FPGA Engineer

Kathmandu, Nepal

Hospo Tech Nepal Pvt. Ltd. (Hotel Yak & Yeti)

July 2019 – April 2021

- Engineered a custom SoPC architecture on **Altera Cyclone II** by integrating a **Nios II/f soft-core processor** via **Avalon-MM fabric**, enabling seamless hardware/software co-design for autonomous solar tracking.
- Designed high-speed **Verilog/VHDL IP cores** for Fuzzy Logic control and Stepper Motor sequencing, offloading compute-intensive routines from the CPU to reduce control-loop latency by **40%**.

Recognition & Awards

Fermilab Technical Excellence Award

2025

Recognized for accelerating Quantinuum testing by 3 months via Python/GPIB test automation and high-quality Verilog deliverables.

Selected Publications

A 32-Channel Cryo-CMOS ASIC for SNSPD Biasing and Readout with Picosecond Timing

Fermilab, NASA, CalTech, 2025

ResearchGate

A 10GHz Low-Jitter Cryogenic Phase-Locked Loop (PLL) For Quantum Applications

Fermilab, Microsoft, 2024

ResearchGate

SPROCKET: Cryogenic 100 KSPS Skipper-CCD Readout IC

IEEE Trans. Nucl. Sci., 2023

IEEE Xplore

Unifying PUF & TRNG over a SAR-ADC

IEEE ISCAS, 2022

IEEE Xplore

Education

University of Illinois at Chicago

Chicago, IL

M.S. in Electrical & Computer Engineering, GPA: 3.75/4.0

Aug 2021 – May 2023

Kamla Nehru Institute of Technology

Sultanpur, India

B.Tech. in Electronics Engineering, GPA: 8.4/10.0

Aug 2015 – May 2019